Structural connectivity predicts individual differences in cognitive outcomes in TBI patients

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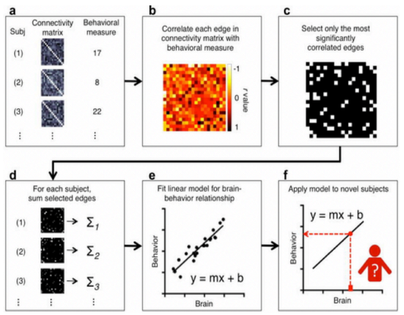
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Introduction

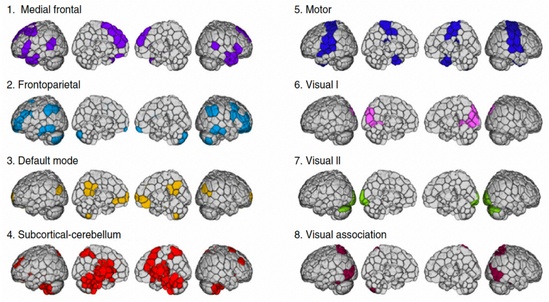
Traumatic brain injury (TBI) patients often suffer from widely distributed axonal damage, named diffuse axonal injury. Although behavioral and cognitive deficit are conspicuous in TBI patients, brain structures and networks that being disrupted by TBI are complicated and widely distributed throughout the brain. The current study aimed at examining if brain structural connectivity between different brain regions can predict TBI patients' cognitive outcome in different cognitive domains.

Method

CPM paradigm



Shen parcellation

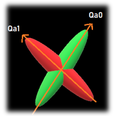


Subjects: 68 TBI and 100 controls (age and sex matched).
Diffusion MRI images reconstructed by GQI (Yeh et. al.2013) in subject's DWI space.
GQI generates ODF for each voxel.

ODF

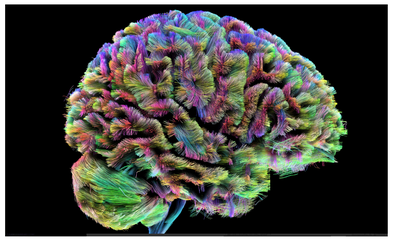


QA (Peaks of ODF)



A whole brain tractography was acquired for each subject and the Shen et al (2013) atlas was transformed into each subject native DWI space before getting a whole brain structural connectivity matrix.
Structural connectivity index:
Mean quantitative anisotropy(QA) (Yeh, 2013).

A whole brain tractography

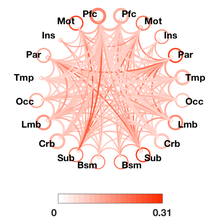


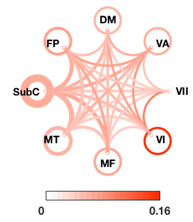
Neuropsychological tests:
Subsets of Wechsler Adult Intelligence Scale (WAIS):
Working memory index(WAISWMI),
Processing Speed index(WAISPSI).
Trail making test A and B.

We explored if structural connectivity can predict cognitive outcomes in TBI patients by applying the connectome based predictive modeling (Finn et al.2015; Rosenberg et al.2016).

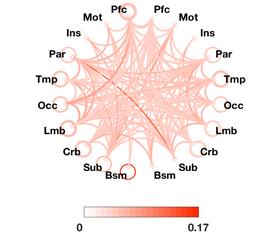
Results

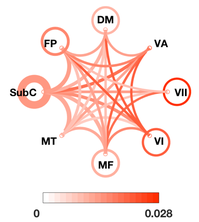
WAISPSI



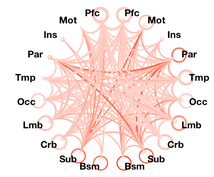


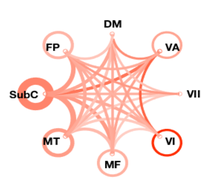
WAISWMI



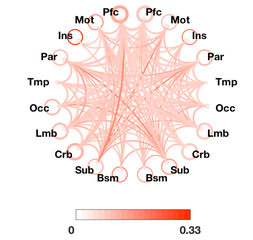


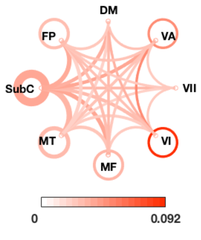
TMT A





TMT B





Canonical networks and anatomical location pairs used in prediction (Hsu et al.2018).
Proportion of total edges of each canonical functional network pair utilized in prediction = number of edges in each network pair included in prediction/the total number of possible edges between that pair of networks.
Line saturation represents the proportion of edges of the particular network pair involved in prediction: the darker the line, the higher the proportion. Line width indicates the total number of edges possible in the pair: the thicker the line, the more possible edges.

Summary

Results are based on 1000 times permutation.
Structural connectivity between prefrontal, motor cortex to subcortical regions successfully predict individual differences in processing speed performance in TBI patients ($r=0.48$, $p<0.0001$).
Structural connectivity between parietal lobes and subcortical regions successfully predict TBI patients' working memory outcomes ($r=0.34$, $p<0.001$).
Structural connectivity between prefrontal and subcortical regions and within the insula predicted cognitive control ability in TBI patients ($r= 0.49$, $p<0.01$).
Our results highlight the role of brain structural connectivity on high-level cognitive functions and the dysconnectivity and damage of different anatomical pathways could result in cognitive impairment in different cognitive domains.

References

1. Finn et al.2015. 2.Rosenberg et al. 2016. 3. Shen et al.2013. 4.Yeh. et al. 2013. 5. Hsu et al. 2018.